

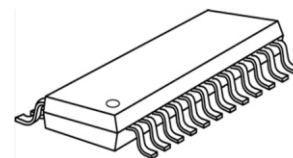


16-Channel PWM Constant Current LED Driver for 1:64Time-Multiplexing Applications

Features

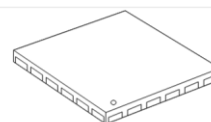
- 3V-5.5V supply voltage
- 16 constant current output channels
- Constant output current range:
 - 0.5~20mA @ $V_{DD}=5V$ supply voltage
 - 0.5~10mA @ $V_{DD}=3.3V$ supply voltage
- Excellent output current accuracy:
 - Between channels: $<\pm 1.5\%$ (Typ.)
 - Between ICs: $<\pm 1.5\%$ (Typ.)
- Built-in 16K-bit x 2 SRAM to support time-multiplexing for 1 ~ 64 scans
- 16/15/14/13 bit color depth PWM control to improve visual refresh rate
- 6-bit current gain, 12.5%~100%
- DCLK double edge trigger
- LED failure isolation
 - LED failure induced cross elimination
- Compulsory LED open detection
- Integrating ghost elimination circuit
- Built-in PWM enhancement
- Improvement of high contrast interference
- GCLK multiplier technology
- Support double refresh mode
- Maximum data clock frequency: 15MHz@ $V_{DD}=5V$

Shrink SOP



GP: SSOP24L-150-0.64

Quad Flat No-leads



GFN: QFN24L-4x4-0.5

Product Description

MBI5264 is designed for LED video applications using internal Pulse Width Modulation (PWM) control with selectable 13-bit / 14-bit / 15-bit / 16-bit color depth. MBI5264 features a 16-bit shift register which converts serial input data into each pixel's gray scale of the output port. Sixteen regulated current ports are designed to provide uniform and constant current sinks for driving LEDs with a wide range of V_F variations. The output current can be preset through an external resistor. The innovative architecture with embedded SRAM is designed to support up to 1:64 time-multiplexing applications. Users only need to send the whole frame data once and to store in the embedded SRAM of the LED driver, instead of sending every time when the scan line is changed. It helps to save the data bandwidth and to achieve high grayscale with very low data clock rate. With scan-type Scrambled-PWM (S-PWM) technology, MBI5264 enhances PWM by scrambling the "on" time of each scan line into several "on" periods and sequentially drives each scan line for a short "on" period. The enhancement equivalently increases the visual refresh rate of scan-type LED displays. In addition, the innovative GCLK multiplier technique doubles visual refresh rate.

MBI5264 drives the corresponding LEDs to the brightness specified by image data. With MBI5264, all output

channels can be built with 16-bit color depth. When building a 16-bit color depth video, S-PWM technology reduces the flickers and improves the image fidelity.

Through compulsory error detection, MBI5264 detects individual LED for open-circuit errors without extra components. MBI5264 equipped an innovative cross elimination function, and it solves the cross phenomenon induced by failure LEDs. Besides, integrated ghost elimination and high contrast interference circuit eases the ghost problems and improve image contrast.

Block Diagram

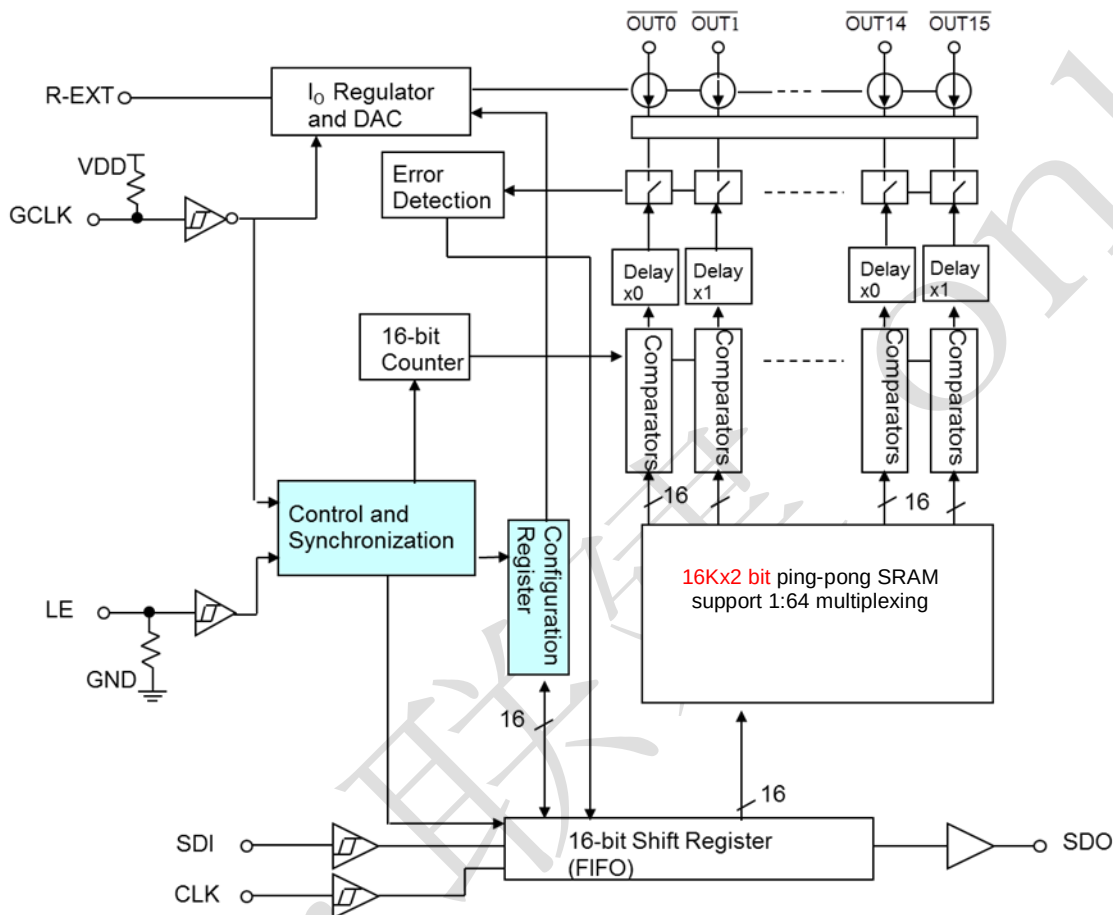
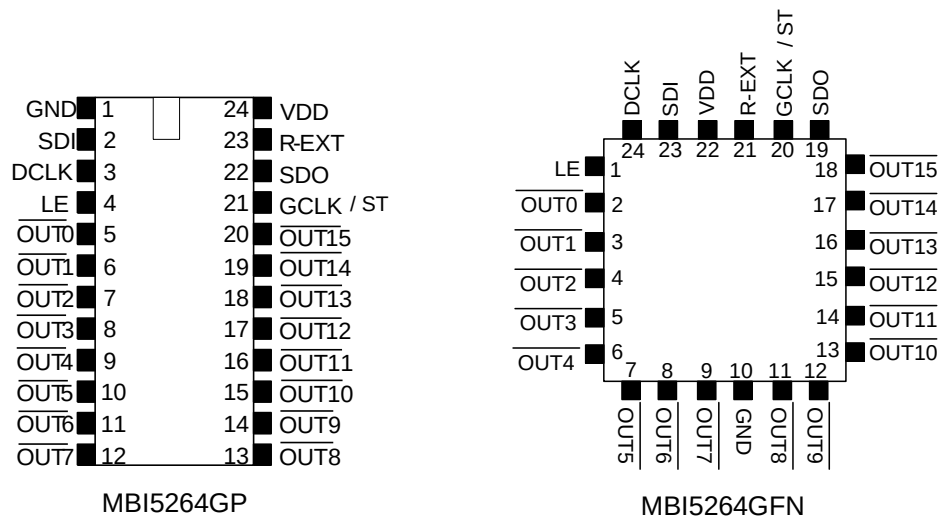


Figure 1

Pin Configuration



Terminal Description

Pin Name	Function
GND	Ground terminal for control logic and current sink
SDI	Serial-data input to the shift register
DCLK	Clock input terminal used to shift data on rising edge and carries command information when LE is asserted.
LE	Data strobe terminal and controlling command with DCLK
OUT0~OUT15	Constant current output terminals
GCLK/ Scan toggle(ST)	external GCLK mode : Gray scale clock terminal Clock input for gray scale. The gray scale display is counted by gray scale clock comparing with input data. internal GCLK mode : Scan toggle terminal Using PLL after driver finishes any scan display, driver will enter deadtime. After external toggle signal changes itself, driver will wait MT internal gclk, and then driver finish deadtime.
SDO	Serial-data output to the receiver-end SDI of next LED driver
R-EXT	Input terminal used to connect an external resistor for setting up output current for all output channels
VDD	3.3V/5V supply voltage terminal

Equivalent Circuits of Inputs and Outputs

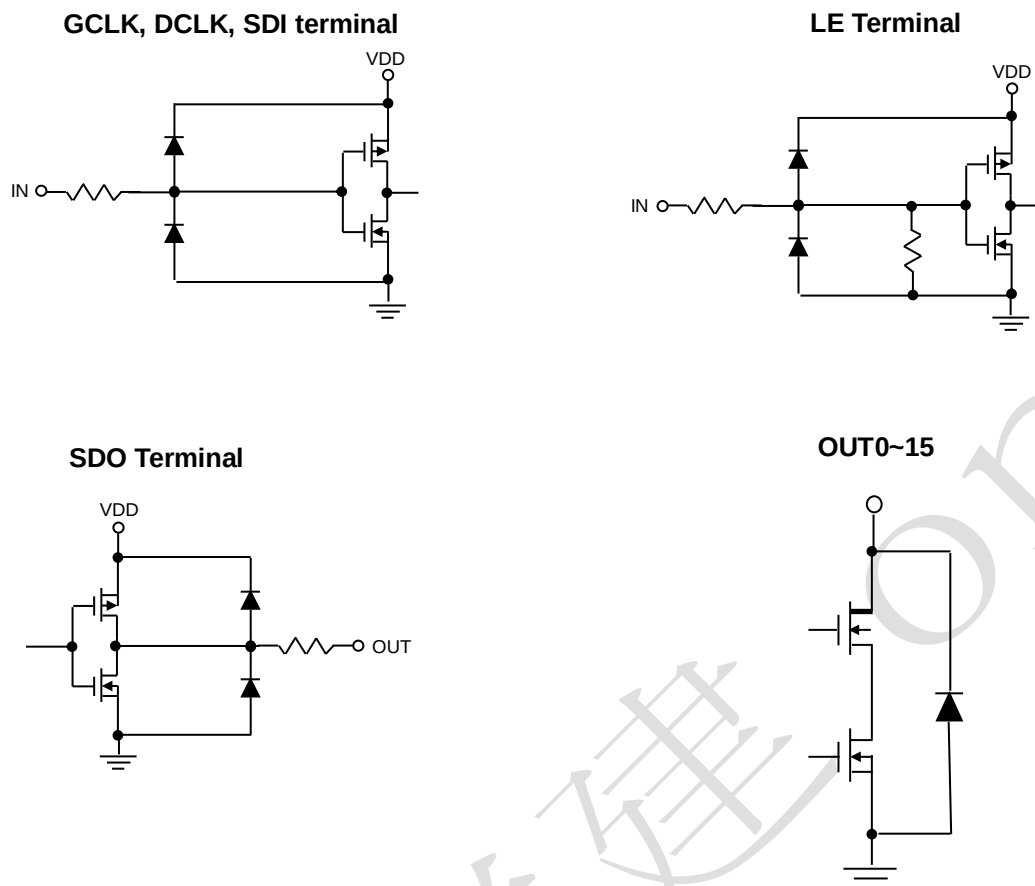


Figure 2

Maximum Rating

Characteristic		Symbol	Rating	Unit
Supply Voltage		V_{DD}	0 ~ 5.5	V
Input Voltage(SDI, DCLK, GCLK, LE)		V_{IN}	-0.4 ~ $V_{DD}+0.4$	V
Output Current per Output Channel		I_{OUT}	+22	mA
Sustaining Voltage at OUT port		V_{DS}	-0.5 ~ $V_{DD}+0.4$	V
GND Terminal Current		I_{GND}	360	mA
Power Dissipation (On 4 Layer PCB, $T_a=25^{\circ}\text{C}$)*	GP	P_D	TBD	W
	GFN		TBD	
Thermal Resistance (On 4 Layer PCB, $T_a=25^{\circ}\text{C}$)*	GP	$R_{th(j-a)}$	TBD	$^{\circ}\text{C/W}$
	GFN		TBD	
Junction Temperature		$T_{j,max}$	150**	$^{\circ}\text{C}$
Operating Temperature		T_{opr}	-40~+85	$^{\circ}\text{C}$
Storage Temperature		T_{stg}	-55~+150	$^{\circ}\text{C}$

*The PCB size is 76.2mm*114.3mm in simulation. Please refer to JEDEC JESD51.

** Operation at the maximum rating for extended periods may reduce the device reliability; therefore, the suggested junction temperature of the device is under 125 $^{\circ}\text{C}$.

Note: The performance of thermal dissipation is strongly related to the size of thermal pad, thickness and layer numbers of the PCB. The empirical thermal resistance may be different from simulative value. User should plan for expected thermal dissipation performance by selecting package and arranging layout of the PCB to maximize the capability.

Electrical Characteristics ($V_{DD}=5.0V$, $T_a=25^{\circ}C$)

Characteristics		Symbol	Condition		Min.	Typ.	Max.	Unit
Supply Voltage		V _{DD}	-		4.5	5.0	5.5	V
Sustaining Voltage at OUT Ports		V _{DS}	OUT0 ~ OUT15		-	-	VDD+0.4	V
Output Current		I _{OUT}	Refer to “Test Circuit for Electrical Characteristics”		0.5	-	20	mA
		I _{OH}	SDO		-	-	-1.0	mA
		I _{OL}	SDO		-	-	1.0	mA
Input Voltage	“H” level	V _{IH}	Ta=-40~85°C		0.7xV _{DD}	-	V _{DD}	V
	“L” level	V _{IL}	Ta=-40~85°C		GND	-	0.3xV _{DD}	V
Output Leakage Current		I _{OH}	V _{DS} =5.4V		-	-	0.5	μA
Output Voltage	SDO	V _{OL}	I _{OL} =+1.0mA		-	-	0.4	V
		V _{OH}	I _{OH} =-1.0mA		4.6	-	-	V
Current Skew (Channel)		dI _{OUT1}	I _{OUT} =0.5mA V _{DS} =1.0V	R _{ext} =14.4kΩ	-	±1.5	±2.5	%
Current Skew (IC)		dI _{OUT2}	I _{OUT} =0.5mA V _{DS} =1.0V	R _{ext} =14.4kΩ	-	±1.5	±3	%
Output Current vs. Output Voltage Regulation*		%/dV _{DS}	V _{DS} within 1.0V and 3.0V,		-	±0.1	±0.3	% / V
Output Current vs. Supply Voltage Regulation*		%/dV _{DD}	V _{DD} within 4.5V and 5.5V		-	±1.0	±2.0	% / V
Pull-down Resistor		R _{IN} (down)	LE			100		KΩ
Supply Current (DCLK=GCLK=0Hz)	“Off”	I _{DD} (off) 1	R _{ext} =Open, OUT0 ~ OUT15=Off		-	2.91	2.96	mA
		I _{DD} (off) 2	R _{ext} =14.4kΩ , OUT0 ~ OUT15=Off		-	2.99	3.04	mA
		I _{DD} (off) 3	R _{ext} =0.72kΩ, OUT0 ~ OUT15=Off		-	4.52	4.57	mA
Supply Current (GCLK=20Hz)	“On”	I _{DD} (on) 2	R _{ext} =14.4kΩ , OUT0 ~ OUT15=ON		-	4.04	4.09	mA
		I _{DD} (on) 3	R _{ext} =0.72kΩ, OUT0 ~ OUT15=ON		-	5.6	5.65	mA

* One channel on.

Electrical Characteristics ($V_{DD}=4.2V$, $T_a=25^{\circ}C$)

Characteristics		Symbol	Condition		Min.	Typ.	Max.	Unit
Supply Voltage		V _{DD}	-		3.8	4.2	4.6	V
Sustaining Voltage at OUT Ports		V _{DS}	OUT0 ~ OUT15		-	-	VDD+0.4	V
Output Current		I _{OUT}	Refer to “Test Circuit for Electrical Characteristics”		0.5	-	20	mA
		I _{OH}	SDO		-	-	-1.0	mA
		I _{OL}	SDO		-	-	1.0	mA
Input Voltage	“H” level	V _{IH}	Ta=-40~85°C		0.75xV _{DD}	-	V _{DD}	V
	“L” level	V _{IL}	Ta=-40~85°C		GND	-	0.25xV _{DD}	V
Output Leakage Current		I _{OH}	V _{DS} =VDD+0.4V		-	-	0.5	μA
Output Voltage	SDO	V _{OL}	I _{OL} =+1.0mA		-	-	0.4	V
		V _{OH}	I _{OH} =-1.0mA		4.6	-	-	V
Current Skew (Channel)		dI _{OUT}	I _{OUT} =0.5mA V _{DS} =1.0V	R _{ext} =14.4kΩ	-	±1.5	±2.5	%
Current Skew (IC)		dI _{OUT2}	I _{OUT} =0.5mA V _{DS} =1.0V	R _{ext} =14.4kΩ	-	±1.5	±3	%
Output Current vs. Output Voltage Regulation*		%/dV _{DS}	V _{DS} within 1.0V and 3.0V,		-	±0.1	±0.3	% / V
Output Current vs. Supply Voltage Regulation*		%/dV _{DD}	V _{DD} within 4.5V and 5.5V		-	±1.0	±2.0	% / V
Pull-down Resistor		R _{IN} (down)	LE			100		KΩ
Supply Current (DCLK=GCLK=0Hz)	“Off”	I _{DD} (off) 1	R _{ext} =Open, OUT0 ~ OUT15=Off		-	10.2	11.5	mA
		I _{DD} (off) 2	R _{ext} =14.4kΩ , OUT0 ~ OUT15=Off		-	12.2	13.5	
		I _{DD} (off) 3	R _{ext} =0.72kΩ, OUT0 ~ OUT15=Off		-	14.2	15.5	
Supply Current (GCLK=20MHz)	“On”	I _{DD} (on) 2	R _{ext} =14.4kΩ , OUT0 ~ OUT15=ON		-	12.2	13.5	
		I _{DD} (on) 3	R _{ext} =0.72kΩ, OUT0 ~ OUT15=ON		-	14.2	15..5	

* One channel on.

Electrical Characteristics ($V_{DD}=3.3V$, $T_a=25^{\circ}C$)

Characteristics		Symbol	Condition		Min.	Typ.	Max.	Unit
Supply Voltage		V _{DD}	-		3.0	3.3	3.6	V
Sustaining Voltage at OUT Ports		V _{DS}	OUT0 ~ OUT15		-	-	VDD+0.4	V
Output Current		I _{OUT}	Refer to “Test Circuit for Electrical Characteristics”		0.5	-	10	mA
		I _{OH}	SDO		-	-	-1.0	mA
		I _{OL}	SDO		-	-	1.0	mA
Input Voltage	“H” level	V _{IH}	Ta=-40~85°C		0.75xV _{DD}	-	V _{DD}	V
	“L” level	V _{IL}	Ta=-40~85°C		GND	-	0.25xV _{DD}	V
Output Leakage Current		I _{OH}	V _{DS} =3.7V		-	-	0.5	μA
Output Voltage	SDO	V _{OL}	I _{OL} =+1.0mA		-	-	0.4	V
		V _{OH}	I _{OH} =-1.0mA		2.9	-	-	V
Current Skew (Channel)		dI _{OUT}	I _{OUT} =0.5mA V _{DS} =1.0V	R _{ext} =14.4kΩ	-	±1.5	±2.5	%
Current Skew (IC)		dI _{OUT2}	I _{OUT} =0.5mA V _{DS} =1.0V	R _{ext} =14.4kΩ	-	±1.5	±3	%
Output Current vs. Output Voltage Regulation*		%/dV _{DS}	V _{DS} within 1.0V and 3.0V,		-	±0.1	±0.3	% / V
Output Current vs. Supply Voltage Regulation*		%/dV _{DD}	V _{DD} within 3.0V and 3.6V		-	±1.0	±2.0	% / V
Pull-down Resistor		R _{IN} (down)	LE			100		KΩ
Supply Current (DCLK=GCLK=0Hz)	“Off”	I _{DD} (off) 1	R _{ext} =Open, OUT0 ~ OUT15=Off		-	9.7	11	mA
		I _{DD} (off) 2	R _{ext} =14.4kΩ , OUT0 ~ OUT15=Off		-	11.7	13	
		I _{DD} (off) 3	R _{ext} =0.72kΩ, OUT0 ~ OUT15=Off		-	13.7	15	
Supply Current (GCLK=20MHz)	“On”	I _{DD} (on) 2	R _{ext} =14.4kΩ , OUT0 ~ OUT15=ON		-	11.7	13	
		I _{DD} (on) 3	R _{ext} =0.72kΩ, OUT0 ~ OUT15=ON		-	13.7	15	

* One channel on.

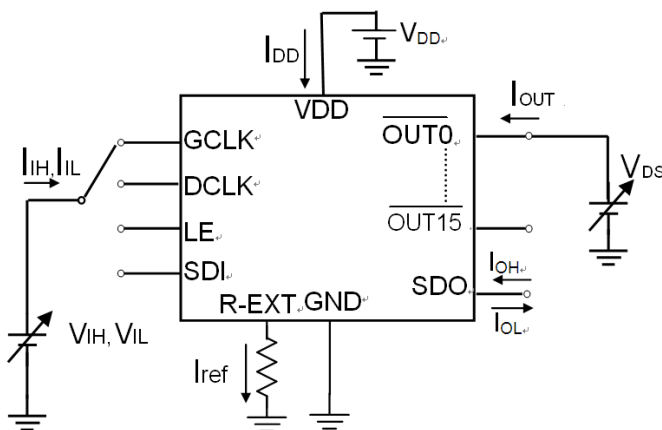
Test Circuit for Electrical Characteristics

Figure 3

for 1:64 Time-multiplexing Applications

Switching Characteristics ($V_{DD}=5.0V$, $T_a=25^{\circ}C$)

Characteristics		Symbol	Condition	Min.	Typ.	Max.	Unit
Setup Time	SDI - DCLK $\uparrow$	t_{SU0}	$V_{DD}=5.0V$ $V_{IH}=V_{DD}$ $V_{IL}=GND$ $R_{ext}=0.72k\Omega$ $V_{DS}=1V$ $R_L=300\Omega$ $C_L=10pF$ $C_1=100nF$ $C_2=10\mu F$ $C_{SDO}=10pF$ $V_{LED}=4.0V$	5	-	-	ns
	LE $\uparrow$ - DCLK $\uparrow$	t_{SU1}		8	-	-	ns
	LE $\downarrow$ (vsync/swrst) - GCLK	t_{SU2}		1200			ns
	LE $\downarrow$ - DCLK $\uparrow$	t_{SU3}^{***}		50			ns
Hold Time	DCLK $\uparrow$ - SDI	t_{H0}		6	-	-	ns
	DCLK $\uparrow$ - LE	t_{H1}		8	-	-	ns
	GCLK - LE $\downarrow$ (vsync/swrst)	t_{H2}		300			ns
Propagation Delay Time	DCLK - SDO	t_{PD0}		-	22	25	ns
	GCLK - $\overline{OUT2n}^*$	t_{PD1}		-	35	-	ns
	LE - SDO	t_{PD2}^{***}		-	30	40	ns
Pulse Width	LE	$t_{W(LE)}$		15			ns
Command to command		t_{CC}		50	-	-	ns
Data Clock Frequency		F_{DCLK}		-	-	30	MHz
Gray Scale Clock Frequency		F_{GCLK}		-	-	33	MHz
Gray Scale Clock Frequency (The function of GCLK doubling is enabled)		F_{GCLK}		-	-	16.6	MHz
Min Clock(GCLK/DCLK) pulse width****		$t_{W(CLK)}$		12	-	-	Ns
Ratio of (GCLK freq)/(DCLK freq)		$R_{(GCLK/DCLK)}$		20	-	-	%
Compulsory error detection operation time****		t_{ERR-C}		700	-	-	ns
Output Rise Time of Output Ports		t_{OR}			15	25	ns
Output Fall Time of Output Ports		t_{OF}			15	25	ns
Dead time positive level		t_{dth}		300			ns
Dead time negative level		t_{dtl}		1200			ns

*Output waveforms have good uniformity among channels.

**In timing of "configuration read", the next DCLK rising edge should be t_{PD2} after LE's falling edge.

***Users have to leave more time than the maximum error detection time for the error detection.

****The Gray Scale Clock period must be 50% duty cycle when the function of GCLK doubling is enabled

Switching Characteristics ($V_{DD}=4.2V$, $T_a=25^{\circ}C$)

Characteristics		Symbol	Condition	Min.	Typ.	Max.	Unit
Setup Time	SDI - DCLK↑	t_{SU0}	$V_{DD}=4.2V$ $V_{IH}=V_{DD}$ $V_{IL}=GND$ $R_{ext}=0.72k\Omega$ $V_{DS}=1V$ $R_L=300\Omega$ $C_L=10pF$ $C_1=100nF$ $C_2=10\mu F$ $C_{SDO}=10pF$ $V_{LED}=4.0V$	5	-	-	ns
	LE↑ - DCLK↑	t_{SU1}		8	-	-	ns
	LE↓(vsync/swrst) - GCLK	t_{SU2}		1200			ns
	LE↓ - DCLK↑	t_{SU3}^{***}		50			ns
Hold Time	DCLK↑ - SDI	t_{H0}		6	-	-	ns
	DCLK↑ - LE	t_{H1}		8	-	-	ns
	GCLK - LE↓(vsync/swrst)	t_{H2}		300			ns
Propagation Delay Time	DCLK - SDO	t_{PD0}		-	22	25	ns
	GCLK - $\overline{OUT2n}^*$	t_{PD1}		-	35	-	ns
	LE - SDO	t_{PD2}^{***}		-	30	40	ns
Pulse Width	LE	$t_{W(LE)}$		15			ns
Command to command		t_{CC}		50	-	-	ns
Data Clock Frequency		F_{DCLK}		-	-	15	MHz
Gray Scale Clock Frequency (The function of GCLK doubling is enabled and only for gclk)		F_{GCLK}		-	-	20	MHz
Gray Scale Clock Frequency (The function of GCLK doubling is disable and only for gclk)		F_{GCLK}		-	-	40	MHz
Min Clock(GCLK/DCLK) pulse width****		$t_{W(CLK)}$		12	-	-	ns
Compulsory error detection operation time***		t_{ERR-C}		700	-	-	ns
Output Rise Time of Output Ports		t_{OR}			15	25	ns
Output Fall Time of Output Ports		t_{OF}			15	25	ns
Dead time positive level		td_{th}		300			ns
Dead time negative level		td_{tl}		1200			ns

*Output waveforms have good uniformity among channels.

**In timing of "configuration read", the next DCLK rising edge should be t_{PD2} after LE's falling edge.

***Users have to leave more time than the maximum error detection time for the error detection.

****The Gray Scale Clock period must be 50% duty cycle when the function of GCLK doubling is enabled

Switching Characteristics ($V_{DD}=3.3V$, $T_a=25^{\circ}C$)

Characteristics		Symbol	Condition	Min.	Typ.	Max.	Unit
Setup Time	SDI - DCLK $\uparrow$	t_{SU0}	$V_{DD}=3.3V$ $V_{IH}=V_{DD}$ $V_{IL}=GND$ $R_{ext}=0.72k\Omega$ $V_{DS}=1V$ $R_L=300\Omega$ $C_L=10pF$ $C_1=100nF$ $C_2=10\mu F$ $C_{SDO}=10pF$ $V_{LED}=4.0V$	7	-		ns
	LE $\uparrow$ - DCLK $\uparrow$	t_{SU1}		10	-		ns
	LE $\downarrow$ (vsync/swrst) - GCLK	t_{SU2}		1200			ns
	LE $\downarrow$ - DCLK $\uparrow$	t_{SU3}^{***}		52			ns
Hold Time	DCLK $\uparrow$ - SDI	t_{H0}		8	-		ns
	DCLK $\uparrow$ - LE	t_{H1}		10	-		ns
	GCLK - LE $\downarrow$ (vsync/swrst)	t_{H2}		300			ns
Propagation Delay Time	DCLK - SDO	t_{PD0}		-	25		ns
	GCLK - $\overline{OUT2n}^*$	t_{PD1}		-	45		ns
	LE - SDO	t_{PD2}^{***}		-	40		ns
Pulse Width	LE	$t_{W(LE)}$		16			ns
Command to command		t_{CC}		52	-	-	ns
Data Clock Frequency		F_{DCLK}		-	-	12.5	MHz
Gray Scale Clock Frequency (The function of GCLK doubling is enabled and only for gclk)		F_{GCLK}		-	-	20	MHz
Gray Scale Clock Frequency (The function of GCLK doubling is disable and only for gclk)		F_{GCLK}		-	-	40	MHz
Min Clock(GCLK/DCLK) pulse width****		$t_{W(CLK)}$		13			ns
Compulsory error detection operation time***		t_{ERR-C}		700	-	-	ns
Output Rise Time of Output Ports		t_{OR}			15	25	ns
Output Fall Time of Output Ports		t_{OF}			15	25	ns
Dead time positive level		t_{dth}		300			ns
Dead time negative level		t_{dtl}		1200			ns

*Output waveforms have good uniformity among channels.

**In timing of "configuration read", the next DCLK rising edge should be t_{PD2} after LE's falling edge.

***Users have to leave more time than the maximum error detection time for the error detection.

****The Gray Scale Clock period must be 50% duty cycle when the function of GCLK doubling is enabled

Test Circuit for Switching Characteristics

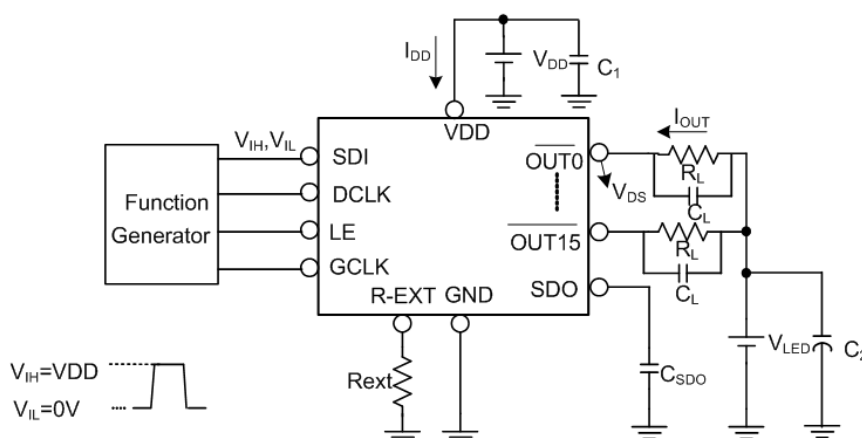
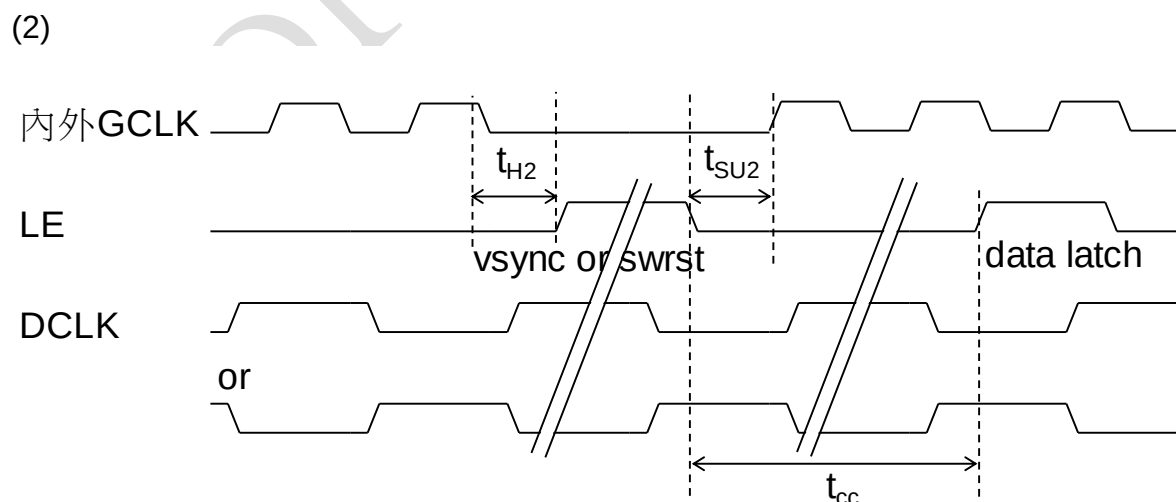
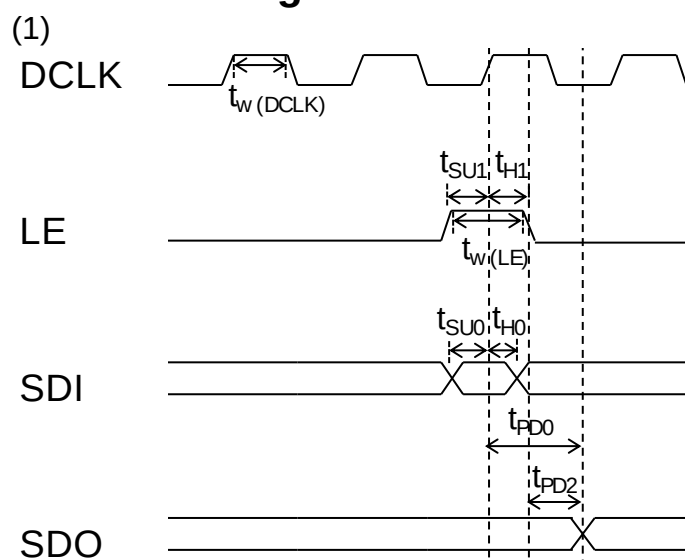


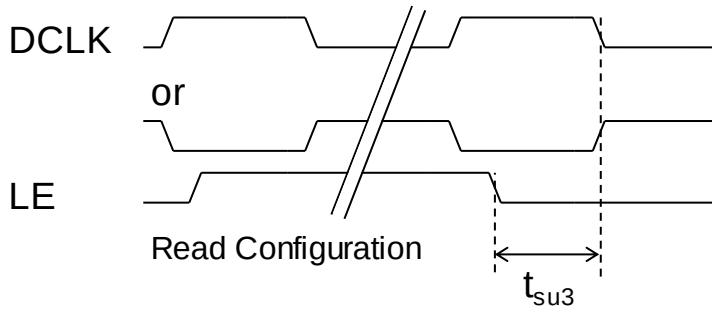
Figure 4

Timing waveform

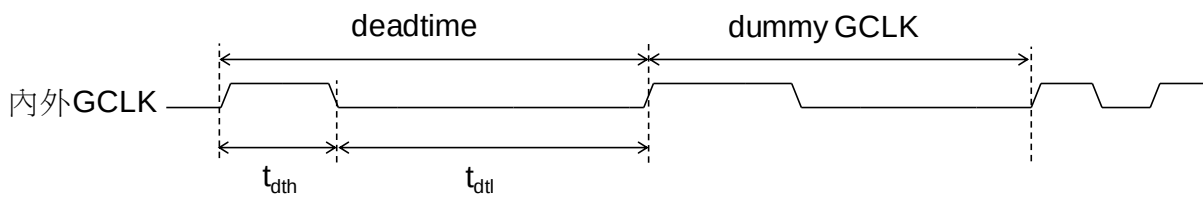
Control timing



(3)



(4)



Control Command

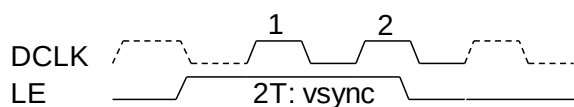
	Signals Combination		Description
Command Name	LE	Number of DCLK Rising Edge when LE is asserted	The Action after a Falling Edge of LE
stop compulsory error detection	High	1	Stop compulsory error detection.
Individual Latch	High	1	Serial data are transferred to the buffers
Vertical Sync	High	2	Display frame will be updated
Compulsory error detection(open error)	High	7	Start compulsory error detection(open error detection)
Software reset	High	10	Reset all the digital part (not including configure registers)
Enable all outputs*	High	11	All 16 channel turn-on
Disable all outputs*	High	12	All 16 channel turn-off
Confirm command	High	14	Confirm command needs to be sent before "write configure, enable all outputs, disable all outputs and test mode" commands.
Write 1 st Configuration*	High	4	Serial data are transferred to the "1 st configuration register"
Write 2 nd Configuration*	High	8	Serial data are transferred to the "2 nd configuration register"
Write 3 rd Configuration*	High	16	Serial data are transferred to the "3 rd configuration register"
Write 4th Configuration*	High	18	Serial data are transferred to the "4th configuration register"
Write 5th Configuration*	High	13	Serial data are transferred to the "5th configuration register"

Write 6th Configuration*	High	15	Serial data are transferred to the “6th configuration register”
Write 7th Configuration*	High	6	Serial data are transferred to the “7th configuration register”
Write 8th Configuration*	High	23	Serial data are transferred to the “8th configuration register”
Write 9th Configuration*	High	24	Serial data are transferred to the “9th configuration register”
Write 10th Configuration*	High	25	Serial data are transferred to the “10th configuration register”
Write 11th Configuration*	High	26	Serial data are transferred to the “11th configuration register”
Write 12th Configuration*	High	27	Serial data are transferred to the “12th configuration register”
Write 13th Configuration*	High	28	Serial data are transferred to the “13th configuration register”
Write 14th Configuration*	High	29	Serial data are transferred to the “14th configuration register”
Write 15th Configuration*	High	30	Serial data are transferred to the “15th configuration register”
Write 16th Configuration*	High	31	Serial data are transferred to the “16th configuration register”
Read 1 st Configuration	High	5	Serial data are transferred from the “1 st configuration register”
Read 2 nd Configuration	High	9	Serial data are transferred from the “2 nd configuration register”
Read 3 rd Configuration	High	17	Serial data are transferred from the “3 rd configuration register”
Read 4th Configuration	High	19	Serial data are transferred from the “4th configuration register”
Read 5th Configuration	High	21	Serial data are transferred from the “5th configuration register”
Read 6th Configuration	High	22	Serial data are transferred from the “6th configuration register”
Read 7th Configuration	High	33	Serial data are transferred from the “7th configuration register”
Read 8th Configuration	High	34	Serial data are transferred from the “8th configuration register”
Read 9th Configuration	High	35	Serial data are transferred from the “9th configuration register”
Read 10th Configuration	High	36	Serial data are transferred from the “10th configuration register”
Read 11th Configuration	High	37	Serial data are transferred from the “11th configuration register”
Read 12th Configuration	High	38	Serial data are transferred from the “12th configuration register”
Read 13th Configuration	High	39	Serial data are transferred from the “13th configuration register”
Read 14th Configuration	High	40	Serial data are transferred from the “14th configuration register”
Read 15th Configuration	High	41	Serial data are transferred from the “15th configuration register”
Read 16th Configuration	High	42	Serial data are transferred from the “16th configuration register”
IC number	High	20	Setting first IC number defined

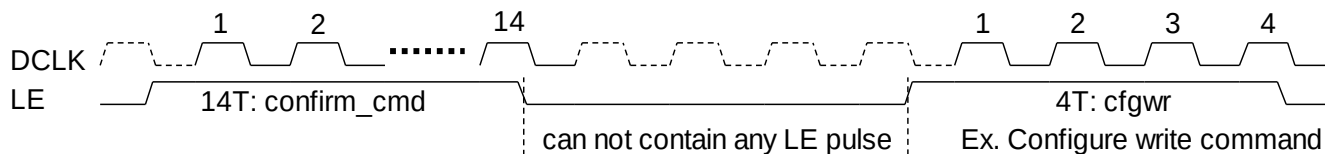
defined			
Write GCLK modify 1 st Configuration*	High	43	Serial data are transferred to the "GCLK modify 1 st configuration register"
Write GCLK modify 2 nd Configuration*	High	44	Serial data are transferred to the "GCLK modify 2 nd configuration register"
Write GCLK modify 3 rd Configuration*	High	45	Serial data are transferred to the "GCLK modify 3 rd configuration register"
Write GCLK modify 4 th Configuration*	High	46	Serial data are transferred to the "GCLK modify 4 th configuration register"
Write GCLK modify 5 th Configuration*	High	47	Serial data are transferred to the "GCLK modify 5 th configuration register"
Read IC information register for gclk modify	High	48	Only for digital test mode Read data format: (Reference to IC information register) IC information register[8:4] = IC number, IC information register[3] = GCLK modify increase or decrease, IC information register[2:0] = GCLK modify value
wakeup	High	62	Force wakeup for sleep mode
HW reset	High	63	Force HW reset active

*Those commands can only be sent after "confirm command".

Command which need no confirm-command ahead



Command which need confirm-command ahead

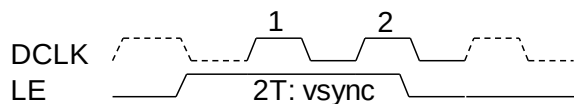


● ind_le



Note: GCLK can not stop during this command, and $GCLK_freq / DCLK_freq \geq 1/5$

● vsync



Definition of 1st Configuration Register

MSB

LSB

F	E	D	C	B	A	9	8	7	6	5	4	3	2	1	0
---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---

e.g. Default Value

F	E	D	C	B	A	9	8	7	6	5	4	3	2	1	0
1	0	0	0	1	1	0	0	1	0	1	1	1	1	1	1

Bit	Attribute	Definition	Value	Function
15	Reserved	Reserved	0~1(default)	Reserved
14	Read/Write	Stripe removal at low greyscale	0(default)~1	0: Disable 1: Enable
13	Read/Write	Cross Effect Removal	0(default)~1	0: Not turn off LED open channel 1: Enable
12~8	Read/Write	Number of scan lines defined ¹	00000 00001 00010 00011(default) ~ 11111	If CFG2[0]=0, CFG1[12:8]= 00000: 1 lines; 00001: 2 lines; 00010: 3 lines; ~ 11110: 31 lines; 11111: 32 lines; If CFG2[0]=1, {CFG7[2], CFG1[12:8]}= 000000: 1 lines; 000001: 2 lines; 000010: 3 lines; ~ 111110: 63 lines; 111111: 64 lines;
7	Read/Write	S-PWM mode (grayscale mode)	0(default) 1	{CFG2[14], CFG1[7]} = 0x00 The 65536 GCLKs(16-bit) PWM cycle is divided into 64 sections, each section has 1024 GCLKs {CFG2[14], CFG1[7]} = 0x01 The 32768 GCLKs(15-bit) PWM cycle is divided into 32 sections, each section has 1024 GCLKs., User still sends 16bit data with 1 bit 0 in LSB bits. Ex., {15'h1234, 1'h0}. {CFG2[14], CFG1[7]} = 0x10 The 16384 GCLKs(14-bit) PWM cycle is divided into 32 sections, each section has 512 GCLKs., User still sends 16bit data with 2bit "0" in LSB bits. Ex., {14'h1234, 2'h0}. {CFG2[14], CFG1[7]} = 0x11 The 8192 GCLKs(13-bit) PWM cycle is divided into 16 sections, each section has 512 GCLKs., User still sends 16bit data with 3bit "0" in LSB bits. Ex., {13'h1234, 3'h0}.
6	Reserved	Reserved	0(default)	Reserved

5~0	Read/Write	Current gain control	000,000 ~ 111,111(default)	000,000: 12.5% ~ 111,111: 100%
-----	------------	----------------------	----------------------------------	--------------------------------------

Default setting of configuration register is 16'h8cbf

Definition of 2nd Configuration Register

MSB

LSB

F	E	D	C	B	A	9	8	7	6	5	4	3	2	1	0
---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---

e.g. Default Value

F	E	D	C	B	A	9	8	7	6	5	4	3	2	1	0
1	0	0	0	0	0	1	0	0	0	0	0	0	0	0	1

Bit	Attribute	Definition	Value	Function
15	Reserved	Reserved	0~1(default)	Reserved
14	Read/Write	S-PWM mode (gray scale mode)	0(default) 1	{CFG2[14], CFG1[7]} = 0x00 The 65536 GCLKs(16-bit) PWM cycle is divided into 64 sections, each section has 1024 GCLKs {CFG2[14], CFG1[7]} = 0x01 The 32768 GCLKs(15-bit) PWM cycle is divided into 32 sections, each section has 1024 GCLKs., User still sends 16bit data with 1 bit 0 in LSB bits. Ex., {15'h1234, 1'h0}. {CFG2[14], CFG1[7]} = 0x10 The 16384 GCLKs(14-bit) PWM cycle is divided into 32 sections, each section has 512 GCLKs., User still sends 16bit data with 2bit "0" in LSB bits. Ex., {14'h1234, 2'h0}. {CFG2[14], CFG1[7]} = 0x11 The 8192 GCLKs(13-bit) PWM cycle is divided into 16 sections, each section has 512 GCLKs., User still sends 16bit data with 3bit "0" in LSB bits. Ex., {13'h1234, 3'h0}.
13~11	Read/Write	Advanced color compensation at low grayscale 2	000(default) ~ 111	000:channel turn ON not shift 001:channel turn ON timing shift 1T gclk 010:channel turn ON timing shift 2T gclk 011:channel turn ON timing shift 3T gclk 100:channel turn ON timing shift 4T gclk 101:channel turn ON timing shift 5T gclk 110:channel turn ON timing shift 6T gclk 111:channel turn ON timing shift 7T gclk Note: backward mode doesn't support Channel turn ON offset
10	Read/Write	Double refresh rate	0 1(default)	0: Double refresh rate ×1 1: Double refresh rate ×2
9	Reserved	Reserved	0~1(default)	Reserved
8~5	Reserved	Reserved	0000(default)	Reserved

4~1	Reserved	Reserved	0000(default)	Reserved
0	Reserved	Reserved	0~1(default)	Reserved

Default setting of configuration register is 16'h8201

Definition of 3th Configuration Register

MSB

LSB

F	E	D	C	B	A	9	8	7	6	5	4	3	2	1	0
---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---

e.g. Default Value

F	E	D	C	B	A	9	8	7	6	5	4	3	2	1	0
1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1

Bit	Attribute	Definition	Value	Function
15	Reserved	Reserved	1(default)	Reserved
14~10	Read/Write	Dark block compensation 1	00000 ~ 11111(default)	0,0001~1,1111: Low Level~High Level 00000 means disable
9~5	Read/Write	Dark block compensation 2	00000 ~ 11111(default)	0,0001~1,1111: Low Level~High Level 00000 means disable
4~0	Read/Write	Dark block compensation 3	00000 ~ 11111(default)	0,0001~1,1111: Low Level~High Level 00000 means disable

Default setting of configuration register is 16'hfff

Definition of 4th Configuration Register

MSB

LSB

F	E	D	C	B	A	9	8	7	6	5	4	3	2	1	0
---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---

e.g. Default Value

F	E	D	C	B	A	9	8	7	6	5	4	3	2	1	0
1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1

Bit	Attribute	Definition	Value	Function
15	Reserved	Reserved	0 1(default)	Reserved
14~10	Read/Write	1 st dim scan line brightness compensation	00000 ~ 11111(default)	0,0001~1,1111: Low Level~High Level 00000 means disable
9~5	Read/Write	Advanced color compensation at low greyscale 1	00000 ~ 11111(default)	0,0001~1,1111: Low Level~High Level 00000 means disable
4~0	Read/Write	Advanced color compensation at low greyscale 3	00000 ~ 11111(default)	0,0001~1,1111: Low Level~High Level 00000 means disable

Default setting of configuration register is 16'hfff

Definition of 5th Configuration Register

MSB

LSB

F	E	D	C	B	A	9	8	7	6	5	4	3	2	1	0
---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---

e.g. Default Value

F	E	D	C	B	A	9	8	7	6	5	4	3	2	1	0
0	1	1	1	0	0	1	0	0	0	0	1	1	0	1	0

Bit	Attribute	Definition	Value	Function
15~8	Reserved	Reserved	01110010	Reserved
7~5	Read/Write	Cross effect removal level	000(default) ~ 111	000~111: Low Level~High Level
4~0	Read/Write	Lower de-ghosting level	00000 11010(default) ~ 11111	0,0001~1,1111: Low Level~High Level 00000 means disable

Default setting of configuration register is 16'h721a

Definition of 6th Configuration Register

MSB

LSB

F	E	D	C	B	A	9	8	7	6	5	4	3	2	1	0
---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---

e.g. Default Value

F	E	D	C	B	A	9	8	7	6	5	4	3	2	1	0
0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Attribute	Definition	Value	Function
15~0	Reserved	Reserved	0100000000000000 00	Reserved

Default setting of configuration register is 16'h4000

Definition of 7th Configuration Register

MSB

LSB

F	E	D	C	B	A	9	8	7	6	5	4	3	2	1	0
---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---

e.g. Default Value

F	E	D	C	B	A	9	8	7	6	5	4	3	2	1	0
1	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0

Bit	Attribute	Definition	Value	Function
15~7	Reserved	Reserved	100000000	Reserved
6	Read/Write	Digital reset	0(default)~1	Manual digital circuit reset
5~3	Reserved	Reserved	0(default)~1	Reserved
2	Read/Write	Number of scan lines	0~1(default)	{CFG7[2], CFG1[12:8]}= 000000: 1 lines; 000001: 2 lines; 000010: 3 lines; ~ 111110: 63 lines; 111111: 64 lines;
1	Reserved	Reserved	00	Reserved

Default setting of configuration register is 16'h8004

Definition of 8th Configuration Register

MSB

LSB

F	E	D	C	B	A	9	8	7	6	5	4	3	2	1	0
---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---

e.g. Default Value

F	E	D	C	B	A	9	8	7	6	5	4	3	2	1	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Attribute	Definition	Value	Function
15~13	Reserved	Reserved	000(default)	Reserved
12	Read/Write	Gradient compensation	0(default)~1	0: Disable 1: Enable
11~5	Reserved	Reserved	0 (default)~1	Reserved
4~0	Read/Write	GCLK change control	00000(default)~11111	00000: without GCLK change 00001: 1T GCLK change 00010: 2T GCLK change 00011: 3T GCLK change ~ 11100: 28T GCLK change 11101: 29T GCLK change 11110: 30T GCLK change 11111: 31T GCLK change

Default setting of configuration register is 16'h0000

Definition of 9th Configuration Register

MSB

LSB

F	E	D	C	B	A	9	8	7	6	5	4	3	2	1	0
---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---

e.g. Default Value

F	E	D	C	B	A	9	8	7	6	5	4	3	2	1	0
0	0	0	0	0	0	0	1	0	0	1	0	0	0	0	1

Bit	Attribute	Definition	Value	Function
15~0	Read/Write	Reserved	0000000100100 001	Reserved

Default setting of configuration register is 16'h0121

Definition of 10th Configuration Register

MSB

LSB

F	E	D	C	B	A	9	8	7	6	5	4	3	2	1	0
---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---

e.g. Default Value

F	E	D	C	B	A	9	8	7	6	5	4	3	2	1	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Attribute	Definition	Value	Function
15~0	Reserved	Reserved	0(default)	Reserved

Default setting of configuration register is 16'h0000

Definition of 11th Configuration Register

MSB

LSB

F	E	D	C	B	A	9	8	7	6	5	4	3	2	1	0
---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---

e.g. Default Value

F	E	D	C	B	A	9	8	7	6	5	4	3	2	1	0
0	0	0	0	0	1	0	0	1	1	0	1	0	1	1	1

Bit	Attribute	Definition	Value	Function
15~10	Reserved	Reserved	000001	Reserved
9~0	Read/Write	Deadtime gclk period	0~ 3~ 215(Default) ~ 1023	Deadtime width= IC internal GCLK period x K 0: K=3 1: K=3 2: K=3 3: K=4 4: K=5 ... 1023: K=1024 Deadtime period must be larger than 1.5us

Default setting of configuration register is 16'h04d7

Definition of 12th Configuration Register

MSB

LSB

F	E	D	C	B	A	9	8	7	6	5	4	3	2	1	0
---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---

e.g. Default Value

F	E	D	C	B	A	9	8	7	6	5	4	3	2	1	0
0	0	0	0	0	0	0	0	1	0	0	0	0	1	0	1

Bit	Attribute	Definition	Value	Function
15~10	Read/Write	Color Compensation at Low Grayscale	000000 (default)~111111	000000: disable 000001: 1x delta extend unit 000010: 2x delta extend unit ... 111111: 63x delta extend unit
9~0	Read/Write	Dummy gclk period	0~3~133(Default)~1023	Dummy width= IC internal GCLK period x N 0: N=3 1: N=3 2: N=3 3: N=4 4: N=5 ... 1023: N=1024

Default setting of configuration register is 16'h0085

Definition of 13th Configuration Register

MSB

LSB

F	E	D	C	B	A	9	8	7	6	5	4	3	2	1	0
---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---

e.g. Default Value

F	E	D	C	B	A	9	8	7	6	5	4	3	2	1	0
1	0	1	1	0	0	1	0	0	0	0	1	1	0	1	1

Bit	Attribute	Definition	Value	Function
15~14	Reserved	Reserved	10(Default)	Reserved
13~7	Read/Write	PLL variable M	0000010~ 1100100(Default) ~ 1111111	2 to 127
6~2	Read/Write	PLL variable N	00010~ 00110(Default)~ 11111	2 to 31
1~0	Read/Write	PLL variable DIV	00~11(Default)	00~11=Div is 00:/8 01:/4 10:/2 11:/1

Default setting of configuration register is 16'hb21b

Definition of 14th Configuration Register

MSB

LSB

F	E	D	C	B	A	9	8	7	6	5	4	3	2	1	0
---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---

e.g. Default Value

F	E	D	C	B	A	9	8	7	6	5	4	3	2	1	0
0	0	0	0	0	0	0	0	0	1	0	0	1	0	0	1

Bit	Attribute	Definition	Value	Function
15~0	Read/Write	Reserved	0000000001001001	Reserved

Default setting of configuration register is 16'h0049

Definition of 15th Configuration Register

MSB

LSB

F	E	D	C	B	A	9	8	7	6	5	4	3	2	1	0
---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---

e.g. Default Value

F	E	D	C	B	A	9	8	7	6	5	4	3	2	1	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Attribute	Definition	Value	Function
15~0	Reserved	Reserved	0(default)	Reserved

Default setting of configuration register is 16'h0000

Definition of 16th Configuration Register

MSB

LSB

F	E	D	C	B	A	9	8	7	6	5	4	3	2	1	0
---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---

e.g. Default Value

F	E	D	C	B	A	9	8	7	6	5	4	3	2	1	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Attribute	Definition	Value	Function
15~0	Reserved	Reserved	0(default)	Reserved

Default setting of configuration register is 16'h0000

Number of Scan Line

MBI5264 supports 1 to 64 scan lines. Please set the configuration register1 bit [C:8] and register7 bit [2] according to the application. The default value '00011' is 4 scan lines.

Gray Scale Mode and Scan-type S-PWM

MBI5264 provides a selectable 16-bit / 15-bit / 14-bit / 13-bit gray scale by setting the configuration register1 bit [7] and register2 bit [14]. The default value is set to '0' for 16-bit color depth. In 14-bit gray scale mode, users should still send 16-bit data with 2-bit '0' in LSB bits. For example, {14'h1234, 2'h0}.

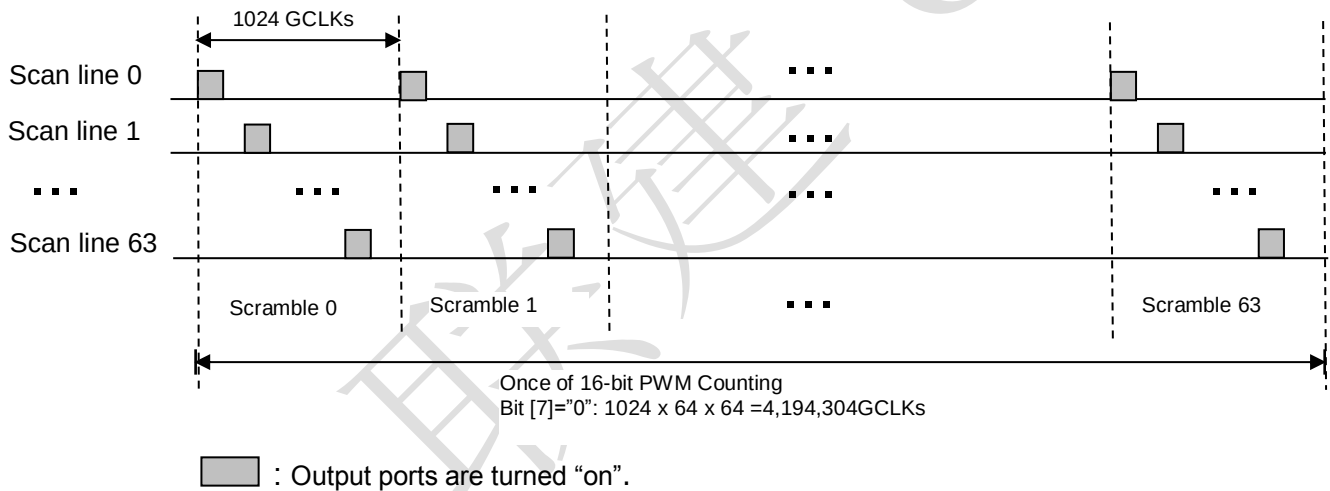
MBI5264 has a smart S-PWM technology for scan type. With S-PWM, the total PWM cycles can be broken into MSB (Most Significant Bits) and LSB (Least Significant Bits) of gray scale cycles. The MSB information can be broken down into many refresh cycles to achieve overall same high bit resolution.

GCLK multiplier

MBI5264 provides a GCLK multiplier function by setting the configuration register1 bit [6]. The default value is set to '0' for GCLK multiplier disable.

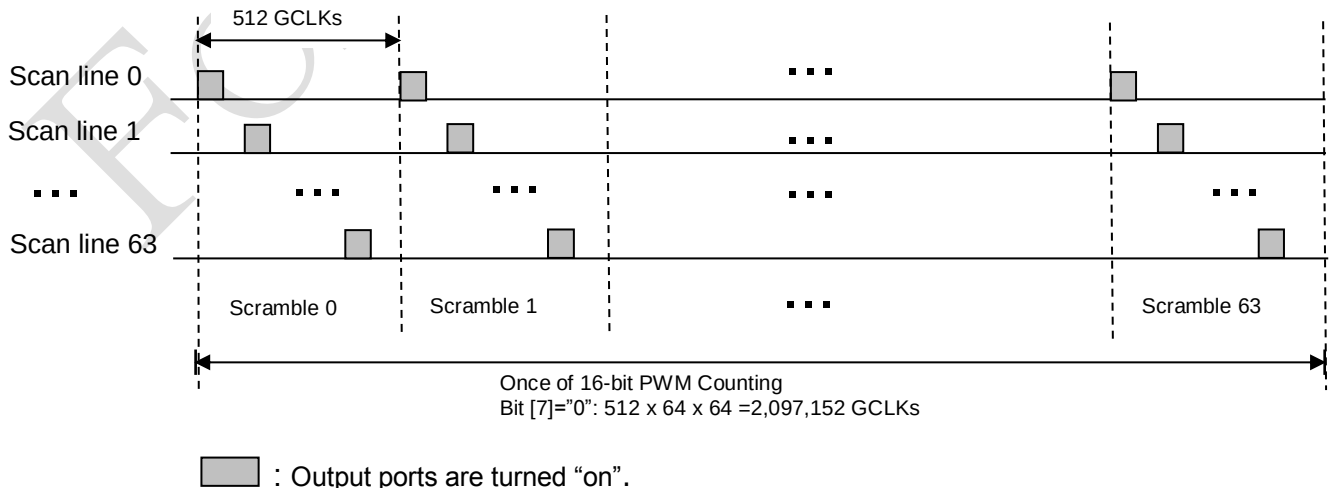
GCLK multiplier disabled (configuration register1 bit [6] = 0)

Display sequence of 64 scrambles



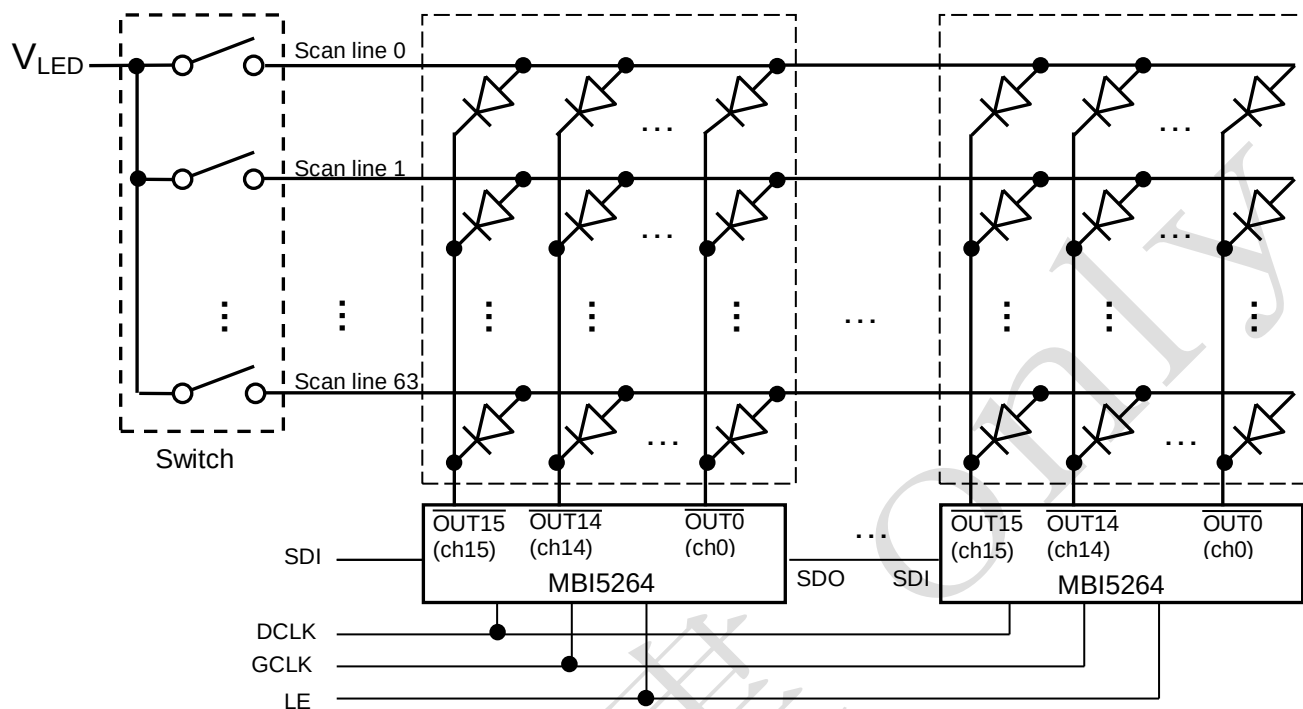
GCLK multiplier enabled (configuration register1 bit [6] = 1)

Display sequence of 64 scrambles



Operation Principal

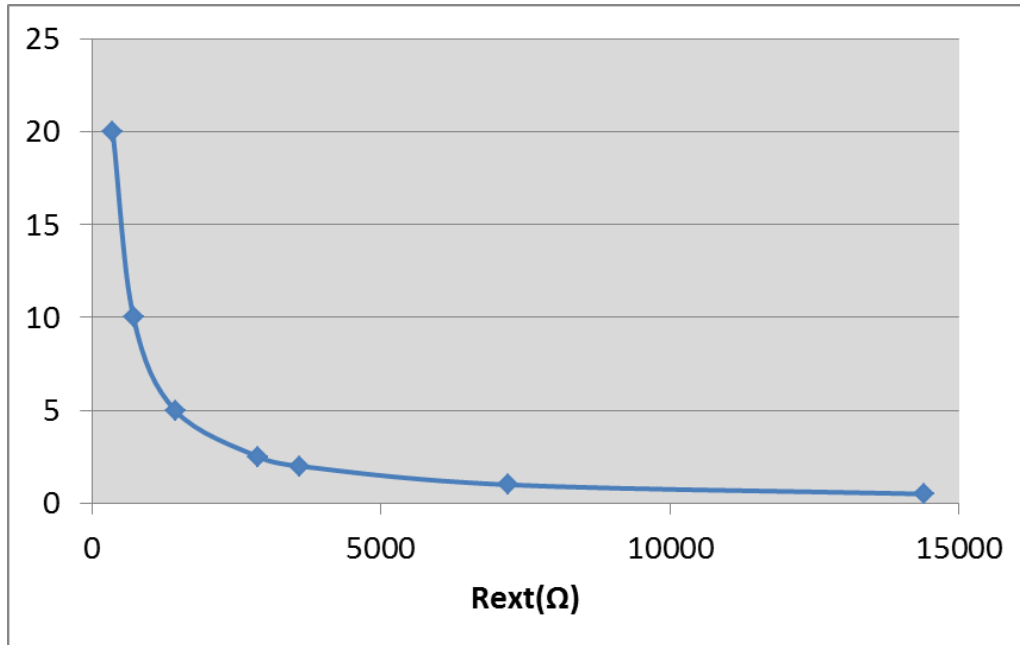
Scan type application structure



The above figure shows the suggested application structure of scan type scheme with 64 scan lines. The gray-scale data are sent by pin “SDI and SDO” with the commands formed by pin “LE” and “DCLK”. The output ports from 16 channels (OUT0 ~ OUT15) will output the PWM result for each scan line at different time, so there must be one “Switch” to multiplex for each scan line. The switching sequence and method and the command usage will be described in the application note.

Setting Output Current

The output current (I_{OUT}) is set by an external resistor, R_{ext} . The default relationship between I_{OUT} and R_{ext} is shown in the following figure.



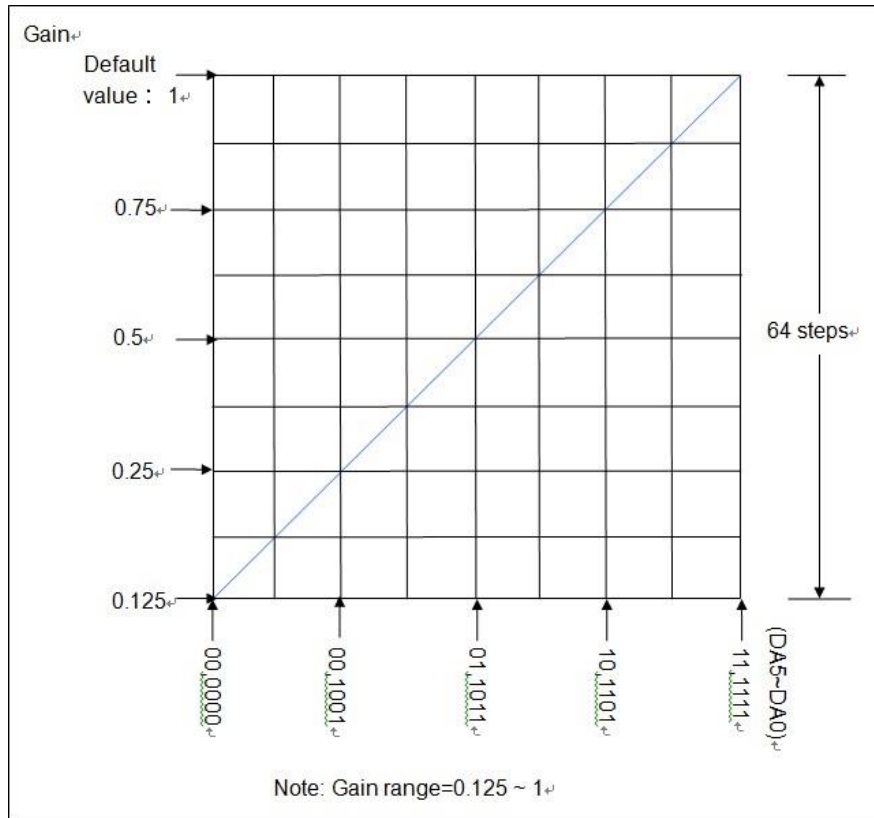
Also, the output current can be calculated from the equation:

$$V_{R-EXT} = 0.3V; I_{OUT} = (V_{R-EXT} / R_{ext}) \times 24 \times G$$

Whereas R_{ext} is the resistance of the external resistor connected to R-EXT terminal and V_{R-EXT} is its voltage. G is the digital current gain, which is set by the bit 5 to bit 0 of the configuration register. The default value of G is 1. For your information, the output current is about 20mA when $R_{ext}=360\Omega$ if G is set to default value 1.

The formula and the setting for G are described in the next section.

Current Gain Adjustment



The 6 bits (bit 5~bit 0) of the configuration register set the gain of output current, i.e., G. As total 6-bit in number, i.e., ranged from 6'b000000 to 6'b111111, these bits allow the user to set the output current gain up to 64 levels.

These bits can be further defined inside configuration register as follows:

F	E	D	C	B	A	9	8	7	6	5	4	3	2	1	0
-	-	-	-	-	-	-	-	-	-	DA5	DA4	DA3	DA2	DA1	DA0

1. Bit 5 to bit 0 are DA5 ~ DA0.

The relationship between these bits and current gain G is:

$$G = 0.125 + (D/63) \times 0.875$$

and D in the above decimal numeration can be converted to its equivalent in binary form by the following equation:

$$D = DA5 \times 2^5 + DA4 \times 2^4 + DA3 \times 2^3 + DA2 \times 2^2 + DA1 \times 2^1 + DA0 \times 2^0$$

In other words, these bits can be looked as 6-bit mantissa DA5~DA0.

For example,

$$G = 0.5, D = (0.5 - 0.125) / 0.875 \times 63 = 27$$

the D in binary form would be:

$$D = 27 = 0 \times 2^5 + 1 \times 2^4 + 1 \times 2^3 + 0 \times 2^2 + 1 \times 2^1 + 1 \times 2^0$$

The 6 bits (bit 5~bit 0) of the configuration register are set to 6'b011011.

Package Power Dissipation (P_D)

The maximum allowable package power dissipation is determined as $P_{D(max)} = (T_J - T_a) / R_{th(j-a)}$. When 16 output channels are turned on simultaneously, the actual package power dissipation is

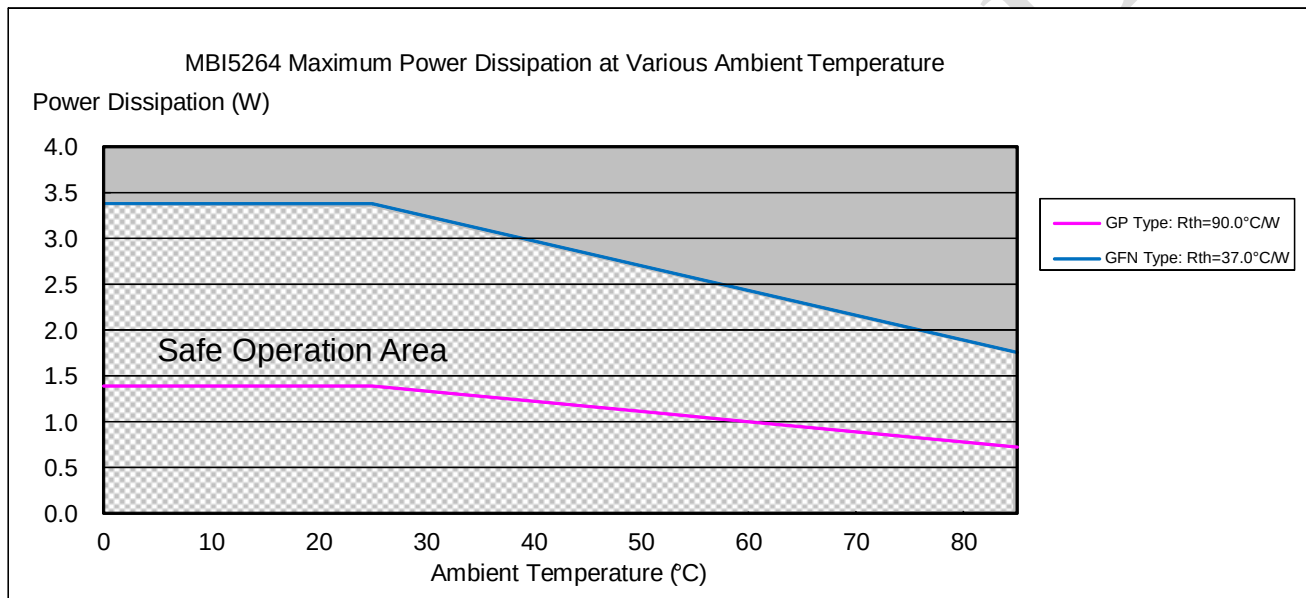
$P_{D(act)} = (I_{DD} \times V_{DD}) + (I_{OUT} \times \text{Duty} \times V_{DS} \times 16)$. Therefore, to keep $P_{D(act)} \leq P_{D(max)}$, the allowable maximum output current as a function of duty cycle is:

$$I_{OUT} = \{[(T_J - T_a) / R_{th(j-a)}] - (I_{DD} \times V_{DD})\} / V_{DS} / \text{Duty} / 16, \text{ where } T_J = 150^\circ\text{C}.$$

Please see the follow table for P_D and $R_{th(j-a)}$ for different packages:

Device Type	$R_{th(j-a)}$ ($^\circ\text{C/W}$)	P_D (W)
GP	90	1.38
GFN	37	3.37

The maximum power dissipation, $P_{D(max)} = (T_J - T_a) / R_{th(j-a)}$, decreases as the ambient temperature increases.



LED Supply Voltage (V_{LED})

MBI5264 is designed to operate with V_{DS} ranging from 0.4V to 1.0V (depending on $I_{OUT}=1\sim 20mA$) considering the package power dissipating limits. V_{DS} may be higher enough to make $P_{D(act)} > P_{D(max)}$ when $V_{LED}=5V$ and $V_{DS}=V_{LED}-V_F$, in which V_{LED} is the load supply voltage. In this case, it is recommended to use the lowest possible supply voltage or to set an external voltage reducer, V_{DROP} .

A voltage reducer lets $V_{DS}=(V_{LED}-V_F)-V_{DROP}$.

Resistors or Zener diode can be used in the applications as shown in the following figures.

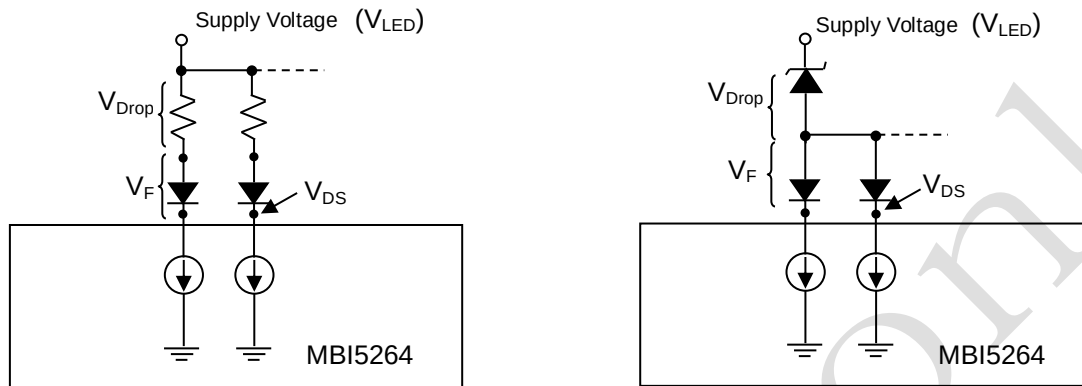


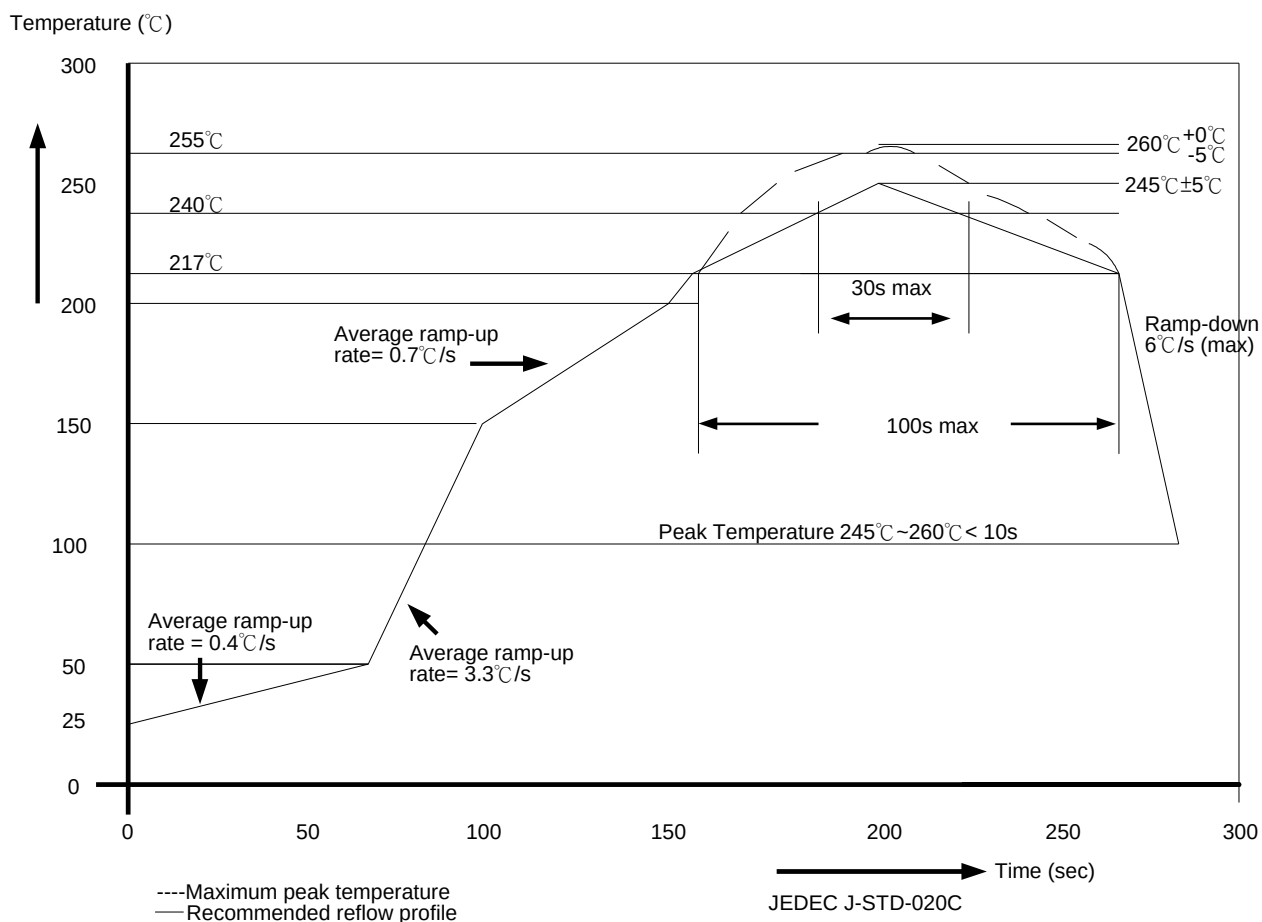
Figure 5

Switching Noise Reduction

LED drivers are frequently used in switch-mode applications which always behave with switching noise due to the parasitic inductance on PCB. To eliminate switching noise, refer to “Application Note for 8-bit and 16-bit LED Drivers- Overshoot”.

Soldering Process of “Pb-free & Green” Package Plating*

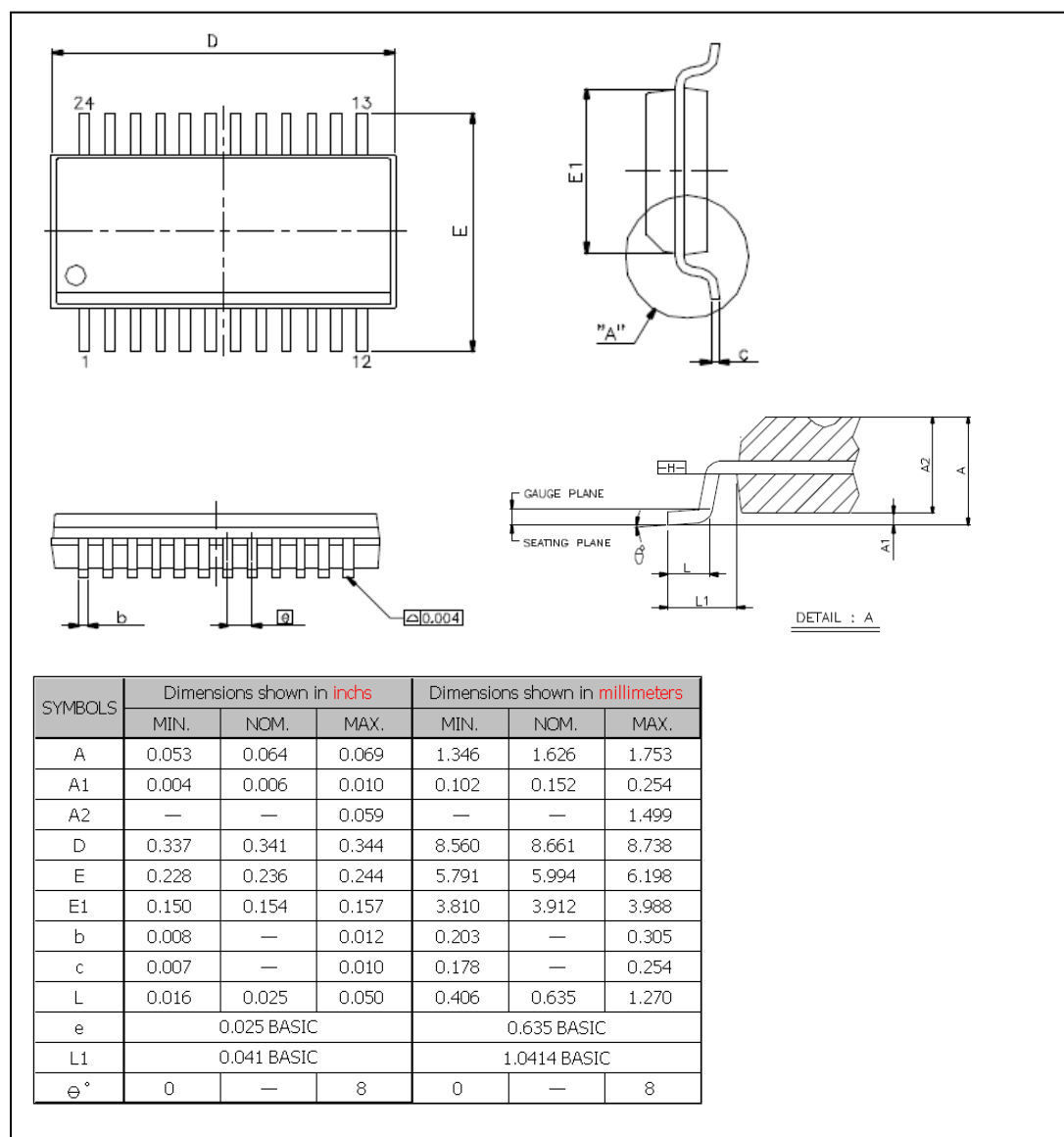
Macroblock has defined "Pb-Free & Green" to mean semiconductor products that are compatible with the current RoHS requirements and selected 100% pure tin (Sn) to provide forward and backward compatibility with both the current industry-standard SnPb-based soldering processes and higher-temperature Pb-free processes. Pure tin is widely accepted by customers and suppliers of electronic devices in Europe, Asia and the US as the lead-free surface finish of choice to replace tin-lead. Also, it adopts tin/lead (SnPb) solder paste, and please refer to the JEDEC J-STD-020C for the temperature of solder bath. However, in the whole Pb-free soldering processes and materials, 100% pure tin (Sn) will all require from 245°C to 260°C for proper soldering on boards, referring to JEDEC J-STD-020C as shown below.



Package Thickness	Volume mm ³ <350	Volume mm ³ 350-2000	Volume mm ³ ≥ 2000
<1.6mm	260 +0 °C	260 +0 °C	260 +0 °C
1.6mm – 2.5mm	260 +0 °C	250 +0 °C	245 +0 °C
≥ 2.5mm	250 +0 °C	245 +0 °C	245 +0 °C

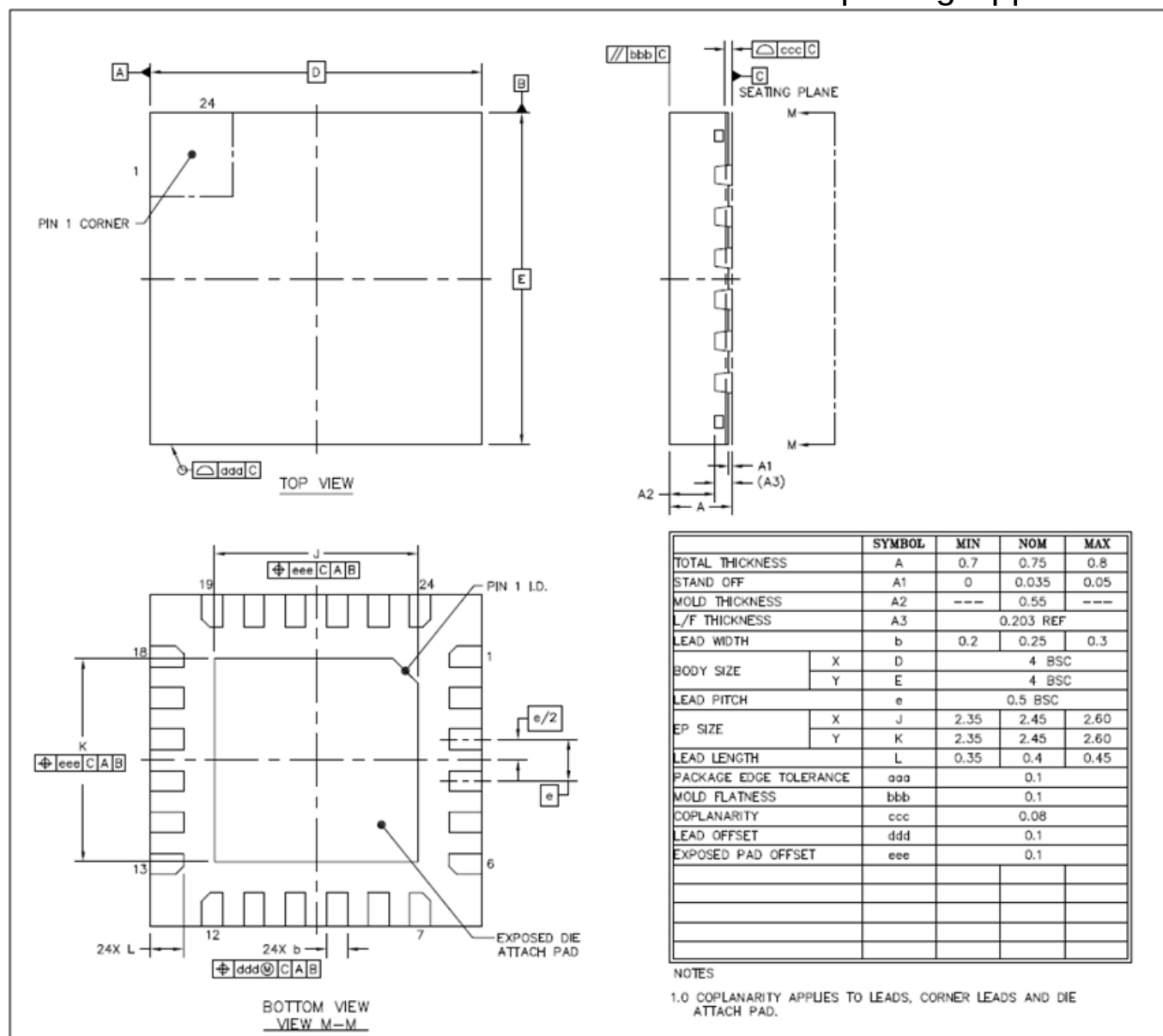
*Note: For details, please refer to Macroblock’s “Policy on Pb-free & Green Package”.

Package Outline



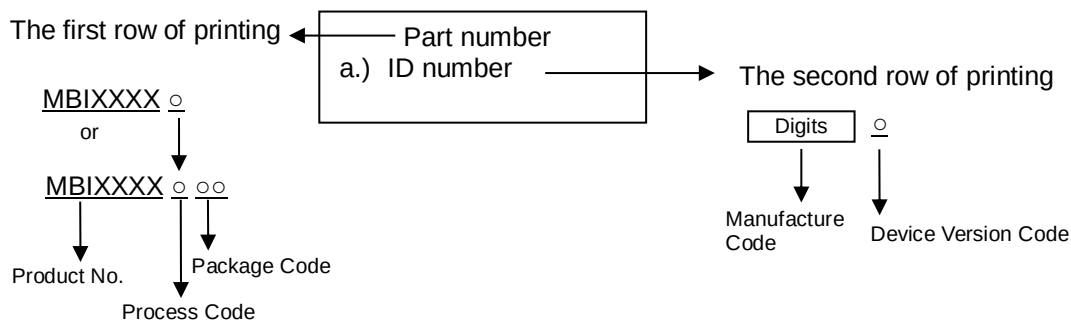
MBI5264 GP Outline Drawing

Note: The unit of the outline drawing is millimeter (mm).



MBI5264GFN Outline Drawing

Note: The unit of the outline drawing is millimeter (mm).

Product Top Mark Information**Product Revision History**

Datasheet Version	Device Version Code
V0.01	T

Product Ordering Information

Product Ordering Number*	RoHS Compliant Package Type	Weight (g)
MBI5264GP-T	SSOP24L-150-0.64	0.11
MBI5264GFN-T	QFN24L-4*4-0.5	0.0379

*Please place your order with the “**product ordering number**” information on your purchase order (PO).

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